Vaibhav Sharma

Machine Learning Engineer

Location: Indiana, USA | **Mail: svaibhav9043@gmail.com** | **Ph:** 877-464-4935| **LinkedIn**

PROFESSIONAL SUMMARY:

4+ years of experience as a Machine Learning Engineer with expertise in NLP (BERT, Hugging Face Transformers, GPT-4, LangChain), recommendation systems (LightGBM, TF-IDF, collaborative filtering), MLOps (Azure ML, Kubernetes, MLflow), data engineering (Apache Spark, Apache Airflow, DBT, Cosmos DB), cloud computing (AWS), data visualization (Tableau, Power BI), and model interpretability (SHAP, LIME, Counterfactual Explanations), driving impactful solutions in finance and AdTech.

TECHNICAL SKILLS:

Programming Languages	Python, SQL, Scala
Machine Learning & AI	TensorFlow, PyTorch, Scikit-Learn, XGBoost, LightGBM, BERT, GPT-4, LangChain
Natural Language Processing (NLP)	Hugging Face Transformers, BERT, LDA, TF-IDF, SHAP, LIME, Counterfactual Explanations
MLOps & Model Deployment	MLflow, Azure ML, Kubernetes, Docker, Prometheus, Grafana
Big Data & Distributed Computing	Apache Spark, Apache Airflow, DBT, HyperLogLog
Cloud Platforms & Services	AWS, Azure, Kubernetes, Cosmos DB
ETL & Data Engineering	Apache Airflow, DBT, ETL Pipelines
Databases & Storage	SQL, NoSQL, Cosmos DB
Data Visualization & Business Intelligence	Tableau, Power BI
A/B Testing & Experimentation	Multi-Armed Bandit Algorithms, A/B Testing Frameworks
Version Control & CI/CD	Git, GitHub, CI/CD Pipelines
Financial & AdTech Domains	Basel III, CCAR, Real-Time Bidding, Lookalike Recommendation Systems

PROFESSIONAL EXPERIENCE:

Charles Schwab – IN | Machine Learning Engineer | February 2025 - Present

- Engineered an NLP-driven sentiment analysis pipeline leveraging Hugging Face Transformers and BERT, improving market prediction accuracy by 35% by extracting sentiment insights from financial news and earnings reports.
- Implemented LLM-based financial text analysis solutions using GPT-4, LangChain and financial embeddings, automating SEC filing analysis, earnings call transcripts and regulatory reports to extract actionable insights.
- Developed cloud-native MLOps pipelines on Azure ML, Kubernetes and MLflow, streamlining model training, versioning and deployment, reducing ML model deployment time by 60%.
- Created dynamic Tableau dashboards integrating real-time market data, risk analytics and trading insights, enabling executives to monitor risk exposure and market trends with live visualizations.
- Automated and optimized data pipelines using Apache Airflow and DBT, improving ETL efficiency by 40%, ensuring seamless data ingestion, transformation and model training for financial analytics.
- Conducted interpretability and explainability analysis using SHAP, LIME and Counterfactual Explanations, ensuring compliance with financial regulations (Basel III, CCAR) and enhancing model transparency for auditability.

Inmobi - India | ML Engineer | Apr 2021 — Jul 2023

- Reduced lookalikes ML recommendation engine end to end execution time by 67% through improved Spark query plan, caching and model checkpointing.
- Engineered augmentation algorithm on top of internal identity data graph to transact extra user IDs to Cosmos DB store, optimizing RUs by 40% and improving CTR by 20%.
- Integrated hyper-log-log data structure in DB storage model and Spark jobs for ETL framework mitigating data availability delay issues and reducing compute costs by 30%.
- Developed and deployed BERT topic modelling LDA model for app description analysis, improving data categorization coherence score by 60%.
- Improved big data pipelines execution framework for incoming data quality monitoring and dashboarding , reducing manual efforts by 50%.
- Analysed ingested user data and implemented forecasting threshold model for fields, resulting in an 80% reduction in data size from order of billions to millions.
- Innovated an ensemble LightGBM and TF-IDF ML model for lookalikes recommendation system resulting in an 20% increase in ROC-AUC and 25% improvement in live metrics.
- Revamped the A/B testing framework implementation and pipelines for performance benchmarking reducing manual efforts by 50%.

Smart AI – India | Machine Learning Engineer | May 2019 - Mar 2021

- Spearheaded the enhancement of the Click-Through Rate (CTR) by 25% through the design and deployment of an advanced recommendation system leveraging collaborative filtering techniques tailored for user engagement in digital advertising.
- Drove a 40% increase in conversion rates by executing rigorous A/B testing frameworks and implementing multi-armed bandit algorithms, leading to data-driven decision-making in ad campaigns.
- Collaborated closely with cross-functional data engineering teams to architect and implement robust ETL pipelines, facilitating efficient data processing and analysis tailored for advertising metrics and insights.
- Employed Apache Spark for high-volume data processing and model training, ensuring scalability and performance optimization in handling advertising datasets.
- Developed and maintained interactive data visualization dashboards using Power BI, enabling real-time tracking of key performance indicators (KPIs) and facilitating strategic decision-making for advertising campaigns.
- Successfully integrated machine learning models into production environments utilizing containerization technologies such as Docker and orchestration tools like Kubernetes, ensuring seamless deployment and scalability.
- Leveraged AWS services for efficient cloud-based deployment of machine learning models and secure data storage, optimizing resource allocation and enhancing operational efficiency.
- Implemented comprehensive monitoring solutions using Prometheus and Grafana to continuously track and evaluate model performance, ensuring adherence to performance benchmarks and timely updates.
- Committed to ongoing professional development by actively engaging in industry research and training programs to stay abreast of the latest advancements in machine learning and trends within the AdTech landscape.

EDUCATION:

Master of Science in Data Science - Indiana University Bloomington, USA
Bachelor of Chemical Engineering - NIT Allahabad, Prayagraj, India